Ancient cuts and modern readjustment: provenance

The ancient anchor is Aristotle's discussion of possible and jointly realised divisions in *On Generation and Corruption* I.2, especially around 317a. The statement that inserting a time node can make neighbouring trajectory vertices readjust is our modern extension. This distinction was clarified in the conversation on 2026-09-10 and is part of the restart context.

1. What the sources supply

Aristotle challenges the inference from divisibility at any chosen point to simultaneous division everywhere, using the absence of immediately consecutive points. Earlier he treats a contact as a relation of two things. These are the direct passage anchors in the Joachim companion.

Plutarch's cone report, section 39, 1079e–1080a, is a companion question about cutting surfaces and neighbouring sections. It reports Democritus's dilemma and Chrysippus's response through Plutarch's critical account. The anywhere-versus-all-together route in I005 came principally from Aristotle, rather than from a statement in the cone passage that inserting a section moves the surrounding sections.

2. What we borrowed and extended

The source-inspired question is whether individually possible cuts can be combined while preserving the compatibility of the whole construction.

In a modern time mesh, inserting y between x and z replaces segment xz by xy and yz. Both adjacent velocity assignments change when y varies. If the old vertices are unknowns in a newly discretised boundary-value or variational problem, their matching equations can change and their solved positions can readjust. This is a modern mechanical interpretation, not an ancient theorem or evidence that Aristotle anticipated path integrals.

Keep three operations distinct:

- Sampling a fixed trajectory adds a record and leaves old positions intact.
- Refining an approximate dynamical construction and solving again can change its old-time vertices. Their readjustment compares mathematical solutions; it does not describe changing the physical past.
- Inserting a variable into an exact consistent path description preserves the old marginal after integration. Conditioning on its observed value can change predictions for correlated variables. Exact classical segment-action composition likewise differs from changing an approximate discretisation.

The one-cut calculation, keeps the two surrounding endpoints fixed. It isolates the local action cost; it does not establish how all free vertices respond to a new global discretisation.

Modern discrete-state and operator comparisons

The earlier conversation's list of discrete-substrate authors was unverified recall, not source evidence for a scientific conclusion. Its useful question is whether loss of classical state information supplies an action restriction. The relevant named source leads are 't Hooft's 1988 equivalence relations, 1999 dissipative system and 2016 book (metadata only). No universal time-grain or information-loss necessity is established from those unread arguments. A finite reversible permutation already has a unitary realization with a self-adjoint logarithm whose spectrum can be bounded below; the phase eigenvalues leave logarithm branches and physical energy normalization undetermined. The former blanket assertion that a reversible automaton has no ground state was too strong.

A discrete tick and a force-dependent record resolution make different claims. The conditional mark bounds neither require a universal time atom nor rule out an independently supplied smaller one. The thermodynamic inequality prices a stipulated information/work trade-off without deriving an absolute floor. The superdeterminism model has its own setting-device mixing and resolution assumptions; it is not a theorem about every deterministic substrate.

Dedekind's cut specifies a real through two rational sets; a finite recorded interval is a different object. The cut-energy proof counts cuts only after a positive record-share floor is supplied. Similarly, writing a spectral-triple differential as $[D, f]$ does not remove points: commutative function algebras still have evaluation states. A distance formula using $\|[D, f]\| \leq 1$ requires its spectral-triple assumptions. Setting $p = \hbar D$ supplies an action conversion rather than deriving one. These are modern mathematical comparisons, not ancient transmission claims.

The groupoid state theorem separates geometric $\epsilon = \hbar/S_0$ from duration and mesh. A mark decision window cannot be declared a distance from its classical fiber by identifying all three parameters. The normalized free blocking test admits both zero and positive action coefficients; it supplies a concrete countertest to automatic scale selection.

Consequence for STATE

Aristotle's anywhere-versus-simultaneous division question, exact sampling, re-solving a discretized boundary-value problem and marginalizing an inserted variable remain distinct. Modern discrete-state and noncommutative definitions add no historical influence or physical action-necessity theorem by analogy. The proofs and measured-source passages, rather than conversation chronology, govern the comparison.